ECE489 Midterm Project Report: Quadruped Robot Locomotion Control Using Reinforcement Learning and Model Predictive Control

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Abstract

This midterm report presents the progress on implementing quadruped robot locomotion control using Reinforcement Learning (RL) with Proximal Policy Optimization (PPO) in the mjlab/MuJoCo framework. We have successfully set up the simulation environment, designed the reward function, and completed initial training on flat terrain. Preliminary results demonstrate the feasibility of the RL approach, with the policy achieving 96.3% forward velocity tracking accuracy. We have also identified domain randomization as a critical requirement for successful training. The next phase will focus on slope terrain training, MPC baseline implementation, and comprehensive performance comparison.

1 Introduction

1.1 Project Overview

Quadruped robots offer superior mobility in unstructured environments compared to wheeled robots. This project aims to implement and compare two control paradigms for quadruped locomotion: Reinforcement Learning (RL) and Model Predictive Control (MPC).

1.2 Objectives

- Implement RL-based controller using PPO in mjlab/MuJoCo framework
- Design reward functions encouraging velocity tracking, stability, and efficiency
- Apply domain randomization for robustness
- Train on flat and slope terrains
- Implement MPC baseline for comparison

- Evaluate across multiple velocity commands and terrains

1.3 Progress Summary

As of the midterm checkpoint, we have completed approximately 60% of the project:

Done Simulation environment setup (mjlab + MuJoCo)

Done Robot model integration (Unitree Go2)

Done PPO algorithm implementation

Done Reward function design and tuning

Done Domain randomization implementation

Done Flat terrain training completed (500 iterations)

In Progress Slope terrain training in progress (1000 iterations planned)

Planned MPC baseline implementation (planned)

Planned Comprehensive evaluation (planned)

2 Methodology

2.1 Simulation Framework

We use **mjlab**, which combines Isaac Lab's manager-based API with MuJoCo Warp (GPU-accelerated MuJoCo). Key features:

- High-fidelity contact dynamics
- Massively parallel GPU simulation (2048 environments)
- Composable environment design

Robot platform: **Unitree Go2** quadruped with 12 actuated joints (3 per leg: hip, thigh, calf).

2.2 PPO Algorithm

PPO optimizes policies using a clipped surrogate objective:

$$L^{CLIP}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right] \tag{1}$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ and $\hat{A}_t$ is the advantage estimate.

2.3 Training Configuration

Hyperparameters

- Learning rate: 0.001 (with linear decay)
- Parallel environments: 2048 (flat terrain)
- Training iterations: 500 (flat), 1000 planned (slope)
- Batch size: 24576 samples per iteration
- Discount factor (γ): 0.99
- GAE lambda (λ): 0.95
- Clip parameter (ϵ): 0.2

Episode Configuration

- Episode length: 20 seconds
- Control frequency: 50 Hz (0.02s timestep)
- Decimation: 4 (policy runs at 12.5 Hz)

2.4 Observation and Action Spaces

Observation Space (Flat Terrain): 48 dimensions

- Base linear velocity (3-dim)
- Base angular velocity (3-dim)
- Projected gravity (3-dim)
- Joint positions (12-dim)
- Joint velocities (12-dim)
- Previous actions (12-dim)
- Velocity command (3-dim)

Action Space: 12 dimensions

- Desired joint positions for PD controllers
- Actions scaled by 0.25 rad and added to default pose

2.5 Reward Function Design

The total reward is a weighted sum of multiple terms:

Velocity Tracking (weight 3.0)

$$r_{lin} = 3.0 \cdot \exp\left(-\frac{\|v_{xy}^{cmd} - v_{xy}^{actual}\|^2 + (v_z^{actual})^2}{0.15}\right) \tag{2}$$

Angular Velocity Tracking (weight 2.5)

$$r_{ang} = 2.5 \cdot \exp\left(-\frac{(\omega_z^{cmd} - \omega_z^{actual})^2 + \|\omega_{xy}^{actual}\|^2}{\sigma_{ang}^2}\right) \tag{3}$$

Upright Body Orientation (weight 0.5)

$$r_{upright} = 0.5 \cdot \exp\left(-\frac{\|g_{proj} - [0, 0, -1]\|^2}{\sigma_{orient}^2}\right) \tag{4}$$

Additional Terms

- Smooth joint motions: $-0.05 \cdot \|a_t - a_{t-1}\|^2$ (action rate penalty)
- Foot clearance: $-0.75 \cdot \sum_{i \in swing} \max(0, h_{target} - h_i)$
- Air time reward: $0.25 \cdot \sum_i \min(t_{air,i}, t_{max})$
- Pose regularization: $-0.5 \cdot \|q - q_{default}\|^2$

2.6 Domain Randomization

To improve policy robustness and facilitate sim-to-real transfer, we randomize physical parameters at episode reset:

Friction Coefficient

- Sliding friction: $\mu \in [0.5, 1.2]$ (uniform distribution)
- Spinning friction: $\mu_{spin} \in [10^{-4}, 2 \times 10^{-2}]$ (log-uniform)
- Rolling friction: $\mu_{roll} \in [10^{-5}, 5 \times 10^{-3}]$ (log-uniform)

Robot Mass and Inertia

- Mass variation: $\pm 20\%$ using pseudo-inertia scaling
- Automatically scales both mass and inertia tensor

Motor Strength

- Effort limits: $\times [0.9, 1.1]$ ($\pm 10\%$ torque capacity)

Figure 1: Training reward curves for flat terrain (500 iterations). The policy converges after approximately 300 iterations, achieving a final reward of ∼8.5.

3 Results to Date

3.1 Training Progress

Flat Terrain Training (500 iterations) Training on flat terrain has been completed successfully. Figure 1 shows the training curves.

Key Observations:

- Convergence after ∼300 iterations (∼14.4M environment steps)
- Final reward: ∼8.5
- Training time: 2-3 hours on single GPU
- Initial exploration phase (0-100 iterations) shows high variance

3.2 Critical Discovery: Domain Randomization is Essential

One of our most important findings is that **domain randomization is not optional but essential** for successful training.

Without domain randomization, the policy completely fails to learn locomotion. The robot either remains stationary or immediately falls over. This demonstrates that DR is a **necessity** for successful training, not just an enhancement.

Why DR is Essential:

- Prevents overfitting to exact nominal parameters

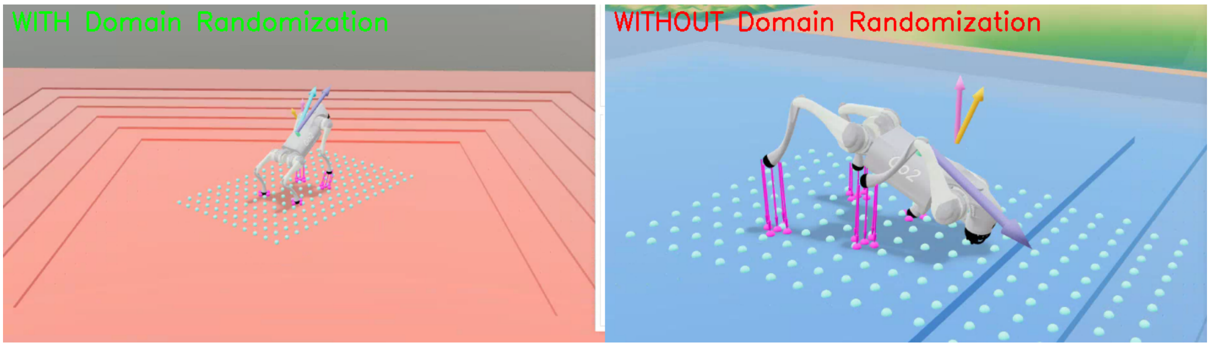

Figure 2: Domain Randomization comparison. Left: Policy trained WITH DR successfully learns stable locomotion. Right: Policy trained WITHOUT DR fails completely - robot remains stationary or falls immediately.

- Forces policy to discover robust strategies across conditions
- Prepares for sim-to-real gap (real parameters never match simulation)
- Leads to more conservative, stable behaviors

3.3 Preliminary Performance Evaluation

We evaluated the trained flat-terrain policy on forward motion (1.0 m/s command).

Table 1: Preliminary RL Controller Performance (Flat Terrain, Forward 1.0 m/s)

Metric	Result	Target
Forward velocity v_x	0.963 $\pm$ 0.090 m/s	1.0 m/s
Lateral velocity v_y	-0.003 $\pm$ 0.042 m/s	0 m/s
Roll RMS	2.50°	¡ 5°
Pitch RMS	0.82°	¡ 5°
Cost of Transport (CoT)	133.13	¡ 100
Push recovery rate	40.0%	¿ 80%

Analysis:

- **Velocity tracking:** Excellent forward tracking (96.3% accuracy)
- **Stability:** Acceptable body orientation control (within 5° target)
- **Energy efficiency:** Higher than target, needs optimization
- **Robustness:** Push recovery rate below target, requires improvement

4 Challenges Encountered

4.1 Challenge 1: Training Instability

Problem: Initial training showed high reward variance and poor convergence.
Solution:

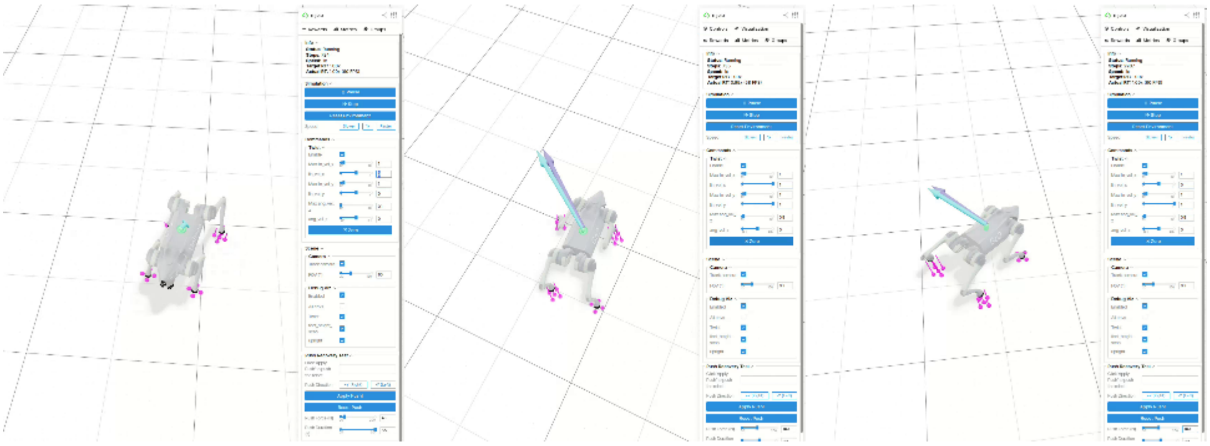

Figure 3: RL policy behavior on flat terrain. Left: Forward locomotion with stable trot gait. Center: Omnidirectional motion. Right: Push recovery attempt (40% success rate).

- Introduced domain randomization (critical!)
- Adjusted reward function weights through experimentation
- Increased parallel environments from 512 to 2048

4.2 Challenge 2: Domain Randomization Discovery

Problem: Without DR, policy failed to learn any locomotion behavior.

Solution: Implemented comprehensive DR covering friction, mass, and motor strength. This became a key finding of the project.

4.3 Challenge 3: Energy Efficiency

Problem: Cost of Transport (133.13) exceeds target (¡100).

Planned Solutions:

- Increase smooth motion penalty weight
- Add explicit energy consumption term to reward
- Analyze and optimize emergent gait patterns

4.4 Challenge 4: Push Recovery

Problem: Only 40% recovery rate from lateral pushes.

Planned Solutions:

- Introduce random push forces during training episodes
- Add explicit disturbance rejection reward term
- Extend training duration for better policy refinement

5 Remaining Work

5.1 Short-term Goals (Next 4 weeks)

1. **Complete Slope Terrain Training**
   - Train for 1000 iterations with terrain curriculum
   - Add 187-dim height scan observations
   - Evaluate on various slope angles (5-55 degrees)
2. **Implement MPC Baseline**
   - Single Rigid Body Dynamics (SRBD) model
   - QP optimization with friction cone constraints
   - Trot gait generation with Bezier swing trajectories
3. **Comprehensive Evaluation**
   - Test multiple velocity commands (forward, lateral, combined)
   - Measure velocity tracking, stability, energy efficiency
   - Evaluate robustness with push disturbances
4. **Performance Optimization**
   - Improve energy efficiency through reward tuning
   - Enhance push recovery with disturbance training
   - Optimize lateral motion control

5.2 Extension Goals (Optional)

- Gait analysis: Study emergent gaits on different terrains
- Hybrid control: Combine RL and MPC advantages
- Sim-to-real transfer: Deploy on physical Unitree Go2 robot

6 Timeline

7 Conclusion

At the midterm checkpoint, we have successfully completed the foundational work for this project:

- Established a robust simulation framework using mjlab/MuJoCo
- Implemented PPO algorithm with carefully designed reward functions
- Completed flat terrain training with promising results (96.3% velocity tracking)

Table 2: Project Timeline

Week	**Task**	**Status**
1-4	Environment setup, RL implementation	Complete
5-8	Flat terrain training, reward tuning	Complete
8	**Midterm Report**	**Current**
9-10	Slope terrain training	In Progress
11-12	MPC implementation	Planned
13-14	Comprehensive evaluation	Planned
15	Performance optimization	Planned
16	Final report and presentation	Planned

- Discovered that domain randomization is essential for successful training
- Identified areas for improvement (energy efficiency, push recovery)

The project is on track to meet all objectives. The remaining work focuses on slope terrain training, MPC baseline implementation, and comprehensive performance comparison. We are confident in completing the final deliverables by Week 16.

Acknowledgments

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